

B.Tech. Degree IV Semester Examination April 2013

CE 402 (A/B) SURVEYING II (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer ALL questions)

(8 × 5 = 40)

- I. (a) What is a compound curve? Write the procedure to set out a compound curve.
 (b) What is a transition curve? Explain the various methods of determining the length of a transition curve.
 (c) Write two systems of co-ordinates employed to locate the position of a heavenly body.
 (d) Explain the following terms:
 (i) Sidereal time (ii) The celestial horizon
 (iii) Sun dial (iv) Altitude (v) The declination
 (e) Explain the term triangulation. What are the classifications of triangulation system?
 (f) Form the normal equations for A, B, C and D in the following equations.
- | <u>Equation</u> | <u>Weight</u> |
|------------------|---------------|
| 3A+4B+3C-D-5 = 0 | 1 |
| A+2B+4C+2D-1 = 0 | 2 |
| 2A+B+2C+5D-2 = 0 | 3 |
- (g) A vertical photograph was taken at an altitude of 1200 meters above mean sea level. Determine the scale of the photograph for terrain lying at elevations of 80 metres and 300 metres if the focal length of camera is 15cm.
 (h) Write a note on the various branches of terrestrial photogrammetry.

PART B

(4 × 15 = 60)

- II. (a) Write the principle and procedure of Rankine's method for setting out a circular curve. (8)
 (b) Determine the offsets to be setout at $\frac{1}{2}$ chain interval along the tangents to locate a 16-chain curve, the length of each chain being 20m (use both methods). (7)
- OR**
- III. Write any three linear methods used for setting out simple circular curve. (15)
- IV. (a) What do you understand by a satellite station? (5)
 (b) Two triangulation stations A & B are 45km apart. Their elevations are 240.00m and 285.00m respectively. The intervening ground may be assumed to have a mean elevation of 222m. Find the minimum height of the signal required to be erected at B so that the line of sight may not graze the ground less than 3m. (10)

OR

(P.T.O.)

- V. (a) Define the term 'most probable value of a quantity. Write the various cases of which the most probable values may be determined. (7)

- (b) The following observations refer to the values of the angles A, B, C at a triangulation station: (8)

$$A = 20^{\circ} 18' 26.4''$$

$$B = 36^{\circ} 28' 14.8''$$

$$C = 56^{\circ} 46' 44.5''$$

Fulfilling the condition that $A + B = C$, find the most probable values of A, B and C .

- VI. Find the declination and hour angle of a star from the following data. (15)

Latitude of the place = $48^{\circ} 30' N$

Azimuth of the star = $50^{\circ} W$

Altitude of the star = $28^{\circ} 24'$

OR

- VII. Find the azimuth and altitude of a star from the following data: (15)

(i) Declination of star = $8^{\circ} 30' S$

(ii) Hour angle of star = 322°

(iii) Latitude of the observer = $50^{\circ} N$

- VIII. What is soundings? Write the different methods of locating soundings. (15)

OR

- IX. (a) Explain the principle of the method of terrestrial photogrammetry. (5)

- (b) A ground area $30\text{km} \times 20\text{km}$ is to be covered by aerial surveying. The following datas are available: (10)

Format size: $230\text{mm} \times 230\text{mm}$

Scale : RF = $1:20,000$

Longitudinal overlapping : 60%

Lateral overlapping : 20%

Calculate the air base length and also the number of photographs required.